

HIERARCHICAL AI-METEOROLOGIST

An LLM-Agent approach for multi-scale and consistent weather forecasting

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ABSTRACT

The Hierarchical AI-Meteorologist is an LLM-agent framework for generating explainable, multi-scale weather reports through hierarchical forecast reasoning. Unlike conventional approaches that model forecasts as flat time series, the proposed system performs structured reasoning across hourly, 6-hourly, and daily temporal scales, capturing both short-term dynamics and longer-term trends. A core reasoning agent transforms structured meteorological inputs into coherent natural-language narratives while extracting a compact set of keywords that summarize dominant meteorological events. These keywords act as semantic constraints for validating temporal coherence and factual consistency. To further ensure causal correctness across scales, the framework incorporates a temporal event graph that explicitly models meteorological event relationships (e. g., pressure drops leading to wind shifts and cooling). Experiments using OpenWeather and Meteostat data demonstrate that hierarchical context and keyword-based validation, augmented by graph-based consistency checking, significantly improve the interpretability and robustness of LLM-generated weather narratives. The proposed framework offers a reproducible approach for semantic evaluation of automated meteorological reporting and advances agent-based reasoning for scientific text generation

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INTRODUCTION

Weather forecasting is the process of predicting atmospheric conditions such as temperature, pressure, wind, and rainfall over time. Traditional systems generate forecasts as flat numerical outputs, making them difficult to interpret across different time scales or explain to non-experts.

Modern approaches use Large Language Models (LLMs) to translate structured meteorological data into natural language reports. However, these models often produce statements that are inconsistent across hourly, six-hourly, and daily scales, and lack mechanisms to verify whether their reasoning actually matches the underlying data.

The Hierarchical AI-Meteorologist addresses this by using a multi-agent LLM framework that reasons across multiple timescales, extracts dominant weather keywords, and validates generated narratives using a causal event graph. This ensures that forecast reports are not only readable, but logically consistent and scientifically grounded.

LITERATURE SUMMARY

- The reviewed literature demonstrates the increasing use of **machine learning, deep learning, and Large Language Models (LLMs)** for weather and climate forecasting.
- **Structured weather datasets** have been shown to improve the quality and validity of forecasting models by converting raw meteorological observations into machine-processable formats.
- **Bi-LSTM models** have achieved strong forecasting performance in structured climate datasets, demonstrating the effectiveness of learning temporal dependencies.
- **CNN-LSTM and other hybrid architectures** combine spatial feature extraction with temporal dependency learning, making them effective for localized temperature forecasting.
- **LSTM-CNN models** have shown strong performance for short-term temperature prediction, while **LSTM-Transformer models** are effective for capturing longer-term seasonal dependencies.
- **TIME-LLM** demonstrates that pre-trained LLMs can be adapted for time-series forecasting without fine-tuning the entire backbone model.
- **CLLMate** shows that combining meteorological raster data with textual knowledge can support forecasting of weather and climate-related events.
- **Weather-R1** highlights the role of logical consistency techniques in reducing contradictory reasoning in weather-related AI systems.

LITERATURE SUMMARY

- **Hierarchical Deep Learning with Temporal Hierarchical Reconciliation** demonstrates the value of maintaining consistency across multiple forecasting timescales.
- Research on modular weather architectures emphasizes the importance of **real-time data integration, scalability, and efficient system design** for operational weather applications.
- The reviewed studies increasingly incorporate **knowledge graphs, structured reasoning, and validation mechanisms** to improve the reliability and interpretability of AI-generated meteorological information.
- Across the studies, the focus is shifting from **pure numerical prediction toward intelligent interpretation, multi-scale forecasting, and natural-language weather reporting**.
- Overall, the literature establishes a strong foundation for combining **weather time-series analysis, hierarchical forecasting, LLM reasoning, and explainable AI** within modern meteorological systems.

RESEARCH GAPS

Inconsistency Across Multiple Timescales

Weather forecasts are generated at different temporal resolutions such as hourly, six-hourly, and daily. Combining these different scales often leads to inconsistencies and fragmented interpretations, as existing systems lack mechanisms to maintain cross-scale coherence.

Incompatibility Between Time-Series Data and Natural Language

Meteorological data is usually stored as dense numerical time-series tables. Translating these structured datasets into clear, human-readable explanations remains challenging because numerical trends must be interpreted and expressed meaningfully in natural language.

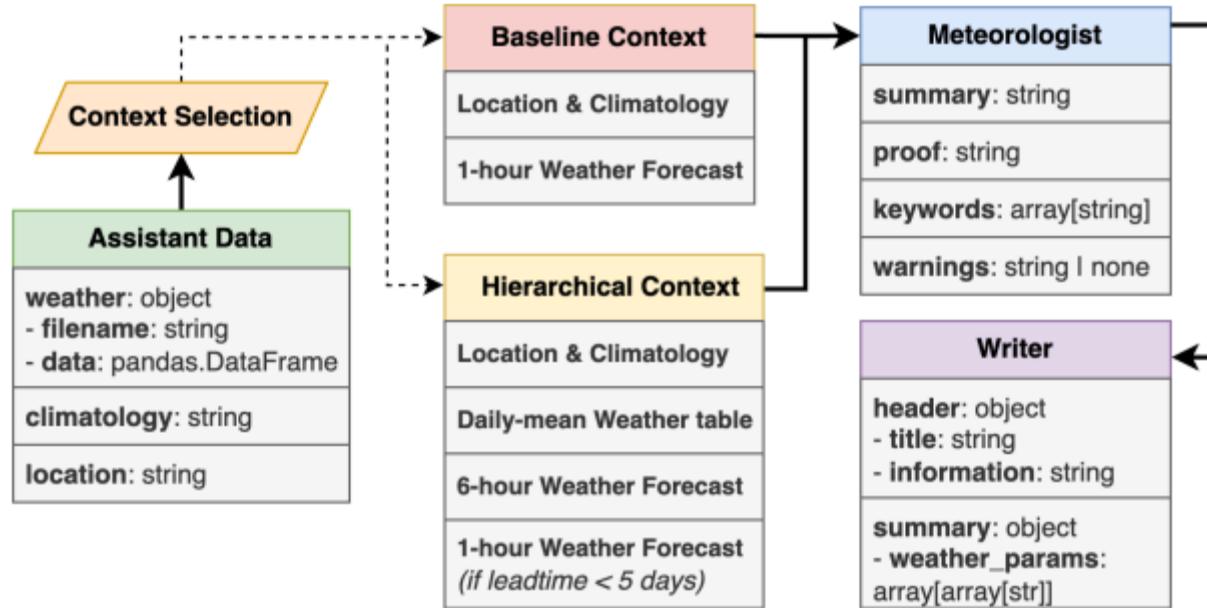
Logical Inconsistency and Lack of Verifiability in LLM Reasoning

Large language models can generate explanations from data but often produce statements that are not explicitly grounded in the underlying numerical values. This makes it difficult to verify whether generated interpretations are logically consistent with the data.

Attention Bias in High-Resolution Long-Horizon Data

Large tables of high-resolution data introduce significant noise, especially over long forecast horizons. Important large-scale patterns or extreme events can become buried among numerous smaller fluctuations, making meaningful trends harder to identify.

EXISTING SYSTEM



From [2]: Hierarchical AI-Meteorologist: LLM-Agent System for Multi-Scale and Explainable Weather Forecast Reporting.

EXISTING SYSTEM

The system has 3 main blocks:

Assistant (non-LLM data processor) Collects and normalizes raw inputs — weather forecasts (OpenWeather), climatology (Meteostat), and location metadata. It aggregates hourly data into 6-hour and daily windows, then packages everything into one of two context modes: Baseline (location + 1-hour forecast) or Hierarchical (location + daily table + 6-hour table + optionally hourly if lead time < 5 days).

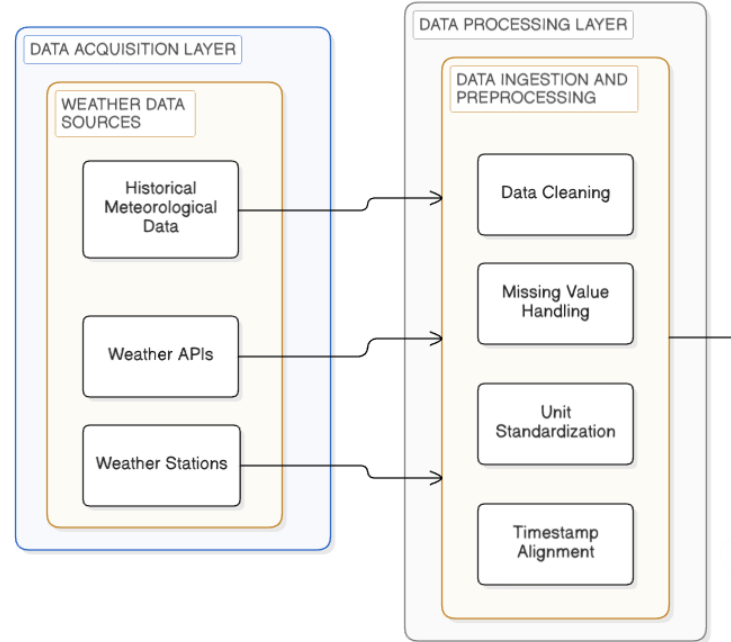
Meteorologist (LLM reasoning agent) Takes the context and produces a structured analysis with 4 fields: summary (multi-paragraph narrative), proof (evidence linking claims to table patterns), keywords (3–5 dominant weather descriptors), and warnings (hazardous conditions if detected).

Writer (LLM style adapter) Receives the Meteorologist's output and adapts tone/format to the target user's domain (e.g., risk analyst, agronomist) without changing any facts. Returns a final JSON report with a header, the full analysis, and an echo of the context tables used.

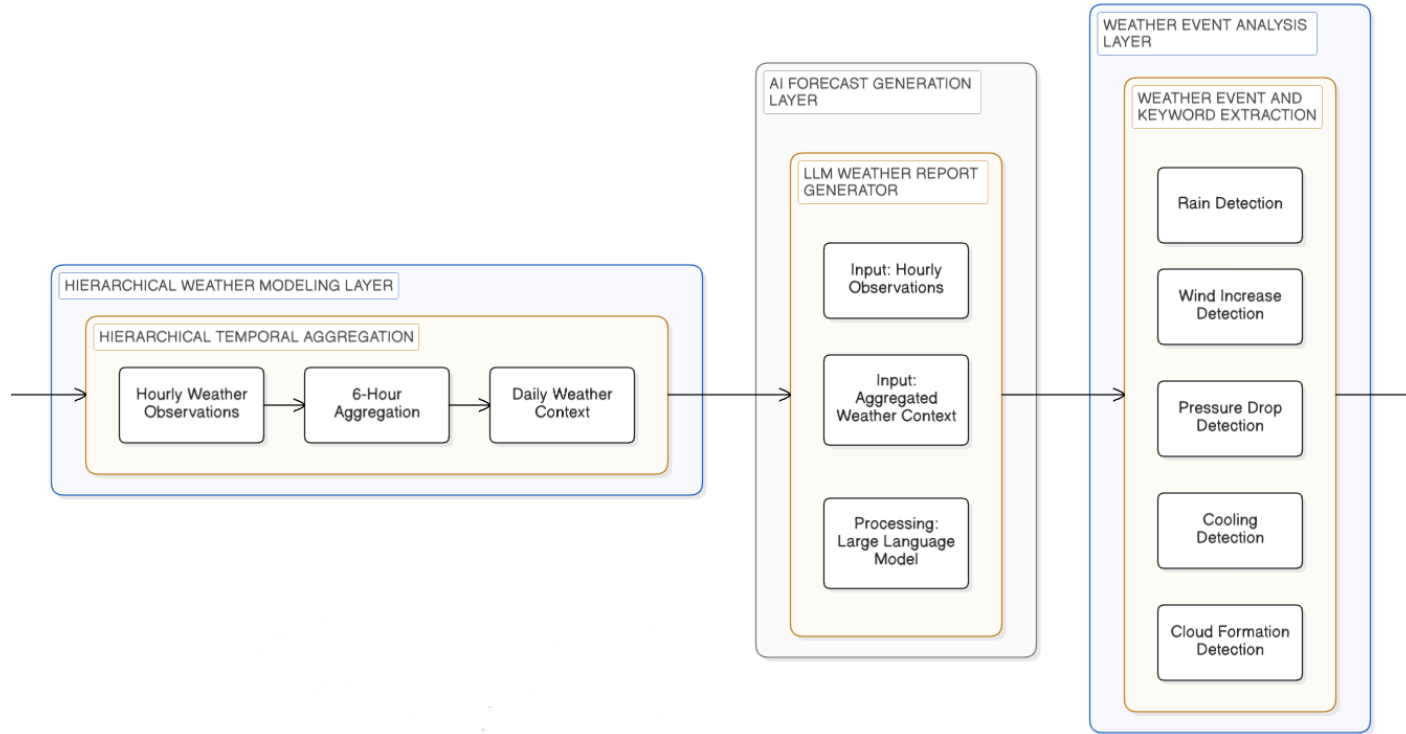
RESEARCH OBJECTIVE

- **Implemented multi-scale weather data processing** by aggregating meteorological information across **hourly, 6-hourly, and daily timescales** to capture both short-term variations and longer-term trends.
- **Implemented meteorological keyword and event extraction** from parameters such as **temperature, pressure, wind, humidity, and rainfall** to summarize dominant weather conditions.
- **Implemented keyword-based and rule-based validation** to check the **logical and temporal consistency** of generated weather narratives across different timescales.
- **Implemented a verification-driven generation pipeline** in which meteorological conclusions are linked to supporting data through **evidence-based Proof Blocks**.
- **Integrated hierarchical context construction with LLM-based reasoning** to generate **consistent, interpretable, and explainable weather reports**.
- **Developed audience-specific report generation** through the Writer Agent while maintaining the factual findings produced by the Meteorologist Agent.

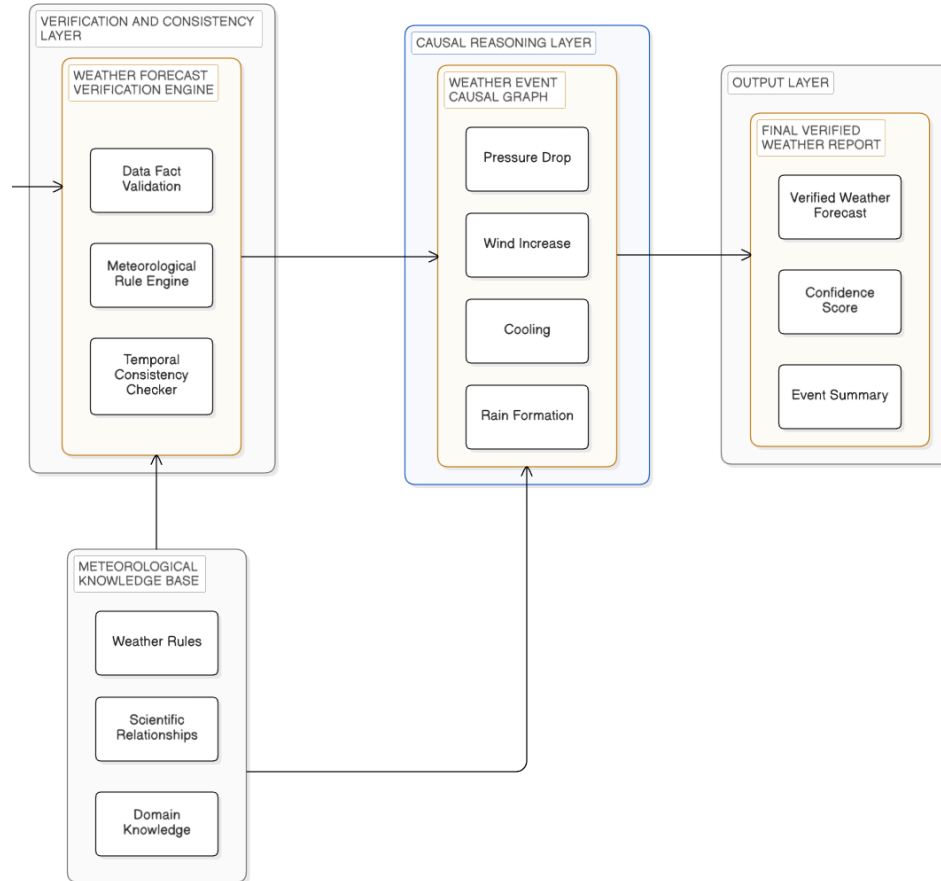
IMPLEMENTED SYSTEM



IMPLEMENTED SYSTEM



IMPLEMENTED SYSTEM



IMPLEMENTATION RESULTS

- Successfully implemented the **three-agent hierarchical meteorological pipeline** with Orchestrator, Meteorologist, and Writer stages.
- The system successfully combines **real-time forecast data, 30-year climatological normals, geospatial metadata, and expert meteorological discussions** into a unified context.
- **Hourly weather data is automatically transformed into 6-hour and daily aggregates**, enabling multi-scale atmospheric analysis.
- The dynamic context mechanism successfully **retains detailed hourly information for short-range forecasts and reduces unnecessary context for longer-range forecasts**.
- The Meteorologist Agent generates **structured atmospheric diagnostics with an evidence-based “Proof” layer**, linking conclusions to underlying weather signals.
- The Writer Agent successfully converts technical analysis into **coherent, audience-specific weather reports** while preserving meteorological fidelity.
- The FastAPI implementation provides **independent access to context generation, meteorological analysis, and final report generation**, enabling modular testing and deployment.

Overall, the implemented system demonstrates **traceable, explainable, and hierarchical weather reasoning rather than simple forecast summarization**.

SOCIAL IMPACT

- Converts complex numerical weather forecasts into **clear and understandable information** for the general public.
- Improves **weather awareness and preparedness** by providing explainable information about upcoming conditions and potential hazards.
- Supports **early decision-making** during events such as heavy rainfall, storms, strong winds, and other extreme weather conditions.
- Helps **local authorities and communities** communicate weather risks more effectively and support timely preventive action.
- Benefits weather-dependent sectors such as **agriculture, transportation, and energy** by making meteorological information easier for non-experts to interpret.
- Promotes **transparent and explainable AI-based weather communication** by linking generated information to underlying meteorological evidence.
- Can improve accessibility to professional weather interpretation for users who may find **technical meteorological data difficult to understand**.

CONCLUSION

The **Hierarchical AI-Meteorologist** demonstrates an explainable multi-agent approach for generating consistent weather reports across **hourly, 6-hourly, and daily timescales**. By combining structured meteorological data, hierarchical temporal aggregation, keyword-based validation, and causal event relationships, the system transforms complex numerical forecasts into coherent and interpretable weather narratives.

The implemented framework addresses key limitations of conventional LLM-based weather reporting, particularly **cross-timescale inconsistency, weak data-to-language grounding, and limited verification of generated reasoning**. The integration of hierarchical context and evidence-based validation provides a more **reliable, traceable, and understandable** approach to automated meteorological reporting.

Overall, the project demonstrates the feasibility of using hierarchical **LLM-agent reasoning** to bridge **numerical weather forecasting** and **explainable natural-language communication**, with potential applications in public safety, agriculture, transportation, and energy.

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